1 The nature of technology: Understanding networks

You will learn

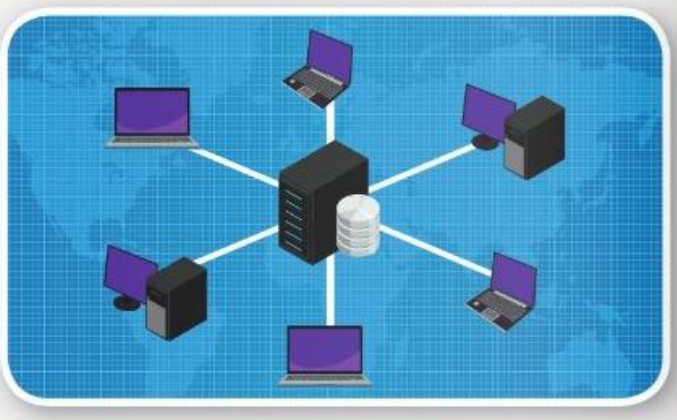
- about network hardware and how the components work together
- how messages are sent over the internet
- about packet switching
- how to connect to a network and how to solve problems with network connections
- about cloud storage and other services in the cloud.

In Student Book 5 you learned how to look for clues that tell you there is a network in your school. You learned that networks use special hardware such as hubs and routers. In this unit you will learn more about networks. You will learn about the hardware components of a network and how they are connected. You will learn how messages are sent over the internet. You will also learn how to connect to a network and how to solve some of the problems you might experience along the way. Finally, you will learn about cloud storage and other services in the cloud.



Unplugged

What do you know about network connections? Look around the room you are in and make a note of everything you can see that tells you there is a network in the room. If there is time, take pictures or make drawings.



Learning outcomes: Explain how computers communicate; Describe internet services (for example, cloud storage)

network hub
switch server router packets
packet switching cloud
data centre wireless access point



Talk about...

In 2019, more than half (around 56%) of the world's population had access to the internet.

The number of people with access to the internet varies in different parts of the world. The rates are high in Europe, North America and parts of Asia. In parts of Africa, access to the internet is still very low.

Is it fair that some people are unable to access the internet? What do you think people miss out on? Is there anything the rest of the world could do to make the internet easier to access?

Did you know?

A new improved mobile phone service called 5G was introduced in 2019. 5G provides an internet connection that is faster than home internet links that run over cables. Some people think that 5G phone networks will become the way that we access the internet at home in the future.

1.1 Network hardware

In this lesson

You will learn:

- about the hardware needed in a network.

Spiral back

In Student Book 5 you learned about two types of network. There are local area networks, like the one in your school, and wide area networks like the internet. Networks use special equipment. You learned how to look for clues that tell you there is a network in your school.

Using networks

When you use a computer, you are usually using a **network**.

You may use your school network. If you are at home, you may have a **broadband network** that connects you to the internet.

When you are sending an email, saving a school assignment or posting your thoughts on a social media site, you use a network. You often don't think about using a network. You save a document and expect it to be available next time you need it. But what happens to the file and where does it go?

When we send or receive a message over a network, we use many pieces of computer hardware. In this lesson you will learn more about network hardware.

Network hardware

Servers and storage

One of the most important pieces of hardware in a network is a **server**. A server is a computer.

The clue to what a server does is in the name. When you go to a restaurant, a waiter serves you food. The job of a waiter is to make sure everyone gets the meal they ordered.



A server stores and sends computer files and messages to users and to devices like printers. The job of a server is to make sure files go to the person or device that they are intended for.

Here are some examples of what a server can do.

It can:

- receive a word-processed file and save it to a network storage drive
- find a spreadsheet file you ask for and send it to your computer
- receive an email from you and pass it on to someone else
- receive a request from you to print a file and send the file to a printer.

Different types of servers do different jobs. For example, there are file servers, print servers and email servers.

Activity

What type of server is used for each of the examples?

What job do you think a web server does?

All computers need to store data. Servers are no different. They have storage devices attached to them. The storage devices used by servers are larger and faster than the ones in personal computers. This is because **network storage** is shared by many people who use the network.

Hubs and switches

When messages are sent out across a network, they need to be sent to the correct place. Hubs and switches do this job. They work in different ways.

When a **hub** receives a message, it sends it to all the computers that are connected to it. Every computer checks the message. The computer that the message is meant for keeps the message. The other computers ignore the message.

When a **switch** receives a message, it decides which computer the message is meant for. It sends the message only to that computer.

So, for example, if a hub is connected to 20 devices, it sends a message to all 20 devices. A switch only sends the message to the device it is meant for.



Hubs and switches: advantages and disadvantages

When a network is busy, the extra messages a hub sends out can slow the network down. A hub is also less secure than a switch. A hacker has more chance of stealing messages.

A switch is more expensive than a hub and is more difficult to set up. But it is more secure and works faster.

Activity

Imagine you work in a computer department. Your manager sends you this email.

I am planning to upgrade the network in our accounts department. The computer manager wants to use a switch for the network. I have been told a hub would be cheaper. What is your advice?

Write a short reply to the email.

Routers and modems



A **router** joins two networks together. Routers are usually used to connect a LAN or a home network to the internet.

When two networks are joined together, they sometimes use different ways to send data. It is like two people talking in different languages. They can communicate if a translator works with them. A translator understands both languages. A translator changes words from one language to the other. This is the job of a **modem** in a network. It takes data sent by one network and translates it into a form that can be understood by the second network.

A wireless access point



Most modern networks allow wireless connection. This means that devices like laptops and tablets can connect to the network without having to use a cable. A wireless connection needs a special piece of hardware called a **wireless access point** (WAP).

A WAP is connected to the network by a cable. But computers can connect to the WAP wirelessly, without a cable. Several computers can connect to a single WAP. WAPs are put on ceilings or high on walls.

A network interface card

A **network interface card** (NIC) allows a computer or other device to connect to a network. Most devices come with a NIC fitted.



A standard desktop or laptop computer will be fitted with two NICs.

- One allows a network cable to be plugged in.
- The other allows a wireless connection.

NICs are fitted inside the computer case but you will see the network cable socket on the outside of the case.



Activity

Look for network hardware at school. Take photos of the hardware you find.

Make a document and label the photos.

If you have a home network, take a photo of your home router and add it to the document.

Extra challenge

Search the web for images of the types of network hardware described in this lesson.

Paste the images at the end of the document you created in the activity. Label them.

Test

1. What does a server do in a network?
2. What piece of network hardware is used to give a wireless connection?
3. How do a router and modem work together in a network?
4. Explain why you would use a switch in a network instead of a hub.

1.2 How a network fits together

In this lesson

You will learn:

- how the hardware components of a network are connected
- how the hardware components work together.

Home network

Many people have internet connections at home. An internet signal reaches your home through telephone or special broadband cables. When the internet connection reaches your home, it is connected to a device called a home router.

In Lesson 1.1 you learned about the devices used in a network. A home router combines the main devices needed for a network into a single small case that sits on a shelf or desk. The small home router contains four main components.

- A **router** connects your home to the internet.
- A **modem** converts the signals that pass along telephone or broadband cables into digital data your home network can use.
- A **switch** makes sure that messages coming into your home are sent to the correct computer.
- A **wireless access point** lets you connect to the network wirelessly, from anywhere in your home.



The photo shows the back of a home router. The yellow cable links the internet to the router. The four empty sockets allow devices to be plugged into the router using a cable. The sockets are sometimes used to connect other devices like a printer.

The three aerials (which look like black sticks in this photo) are connected to the wireless access point. Not all WAPs have aerials, but aerials can improve the wireless signal. Most people connect computers to the router using a wireless connection.

Activity

Which components of a network listed in Lesson 1.1 are not included in a home router? List them.

Explore more

If you have a home broadband connection, examine the home router carefully. Can you identify the features described in this lesson?

Don't touch or remove any cables. It is an electrical device so can be dangerous.

Local area network

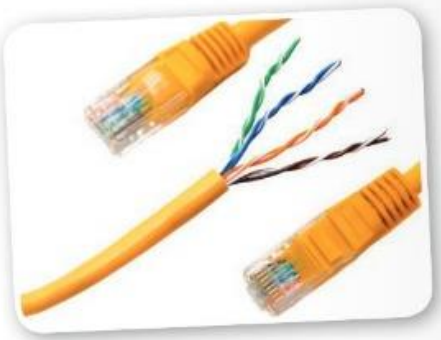
A **local area network** (LAN) is much bigger than a home network, but it contains the same components. The components in a LAN are larger and there are many more of them. A LAN is not stored in a single case like a home network. LAN equipment is spread around a building. The equipment is stored in cabinets that either stand on the floor or are attached to walls.

Connecting LAN components

Network components like servers and switches are connected by cables.

Two types of cable connect network equipment.

- **Copper cable** is the most common cable in a network. Networks use **twisted pair cables**, made up of pairs of thin copper wires twisted together. Data is sent along copper cables as pulses of electricity.
- **Fibre-optic cable** is made up of thin strands of clear fibre. Data is sent along **fibre-optic cables** as pulses of light.





There are three important differences in the two types of cable.

1. More data can be sent at faster speed along fibre-optic cables. Data is sent as light. Nothing travels faster than light.
2. Fibre-optic cables can be used over longer distances than copper. Electrical pulses fade as they travel. Copper cables can only be used over distances of 100 metres.
3. It is cheaper to use copper cable.

The server room

At the centre of a network is a **server room**. The room is usually air conditioned. This is because computer equipment makes a lot of heat. If it gets too hot, the equipment may break down.

The server room contains all the servers in the network. It also contains hubs and switches so that the rest of the network can be connected to the servers.

All servers must be connected to a hub or switch. Data passes along a single cable between the server and a hub or switch. All devices must be connected to a hub or switch. A single hub or switch can connect to many devices. Each connection uses its own cable.

When a computer is plugged into a network socket, it is connected to the network. It sends messages along a cable to the hub or switch. The hub or switch passes the messages from many devices to and from a server.



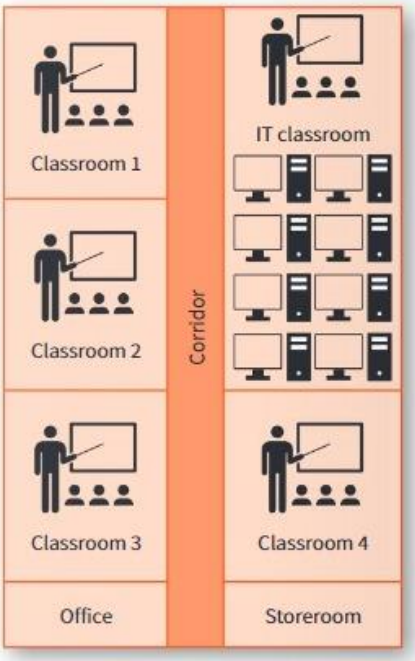
The hub room

Sometimes you need to connect an area of a building to a network that is more than 100 metres away from the server room. This can't be done with copper cable. The electrical signal won't transmit that far. So a **hub room** is set up close to the area that is going to be connected to the network.

A hub room does not contain servers. It only has switches and hubs. Every network socket in the distant area is connected to a hub or switch in the hub room. The hubs or switches in the hub room are connected to the main server room using a fibre-optic cable. Using a fibre-optic cable means that the network can be more than 100 metres from the server room.

Activity

The diagram shows an area in a school that is going to be connected to the school LAN. The area contains four classrooms, an IT classroom containing 20 computers, an office and a storeroom.



Each of the classrooms (including the IT classroom) must have:

- two network connections on the teacher's desk
- a network connection for a WAP installed on the ceiling.

The IT classroom should also have:

- a single network connection for each of the 20 computers
- a network connection for a printer.

The office will be used by four teachers. Each will have their own desk.

- Each desk will have two network connections.
- There will be an extra network connection for a printer in the office.

Draw a copy of the plan. In each room, write the number of network connections needed.

How many network connections are needed in total?

Extra challenge

Hubs or switches for the network will be kept in a cabinet on the wall of one of the rooms. Suggest two rooms that the cabinet could go in.

The school has two spare switches it can use. Each switch has 24 input connections. Does the school need to buy any more switches?

Test

1 List four network components that are contained in a home router.
2 What are the two types of cable used in LANs?
3 Give an example of where a fibre-optic cable is used in a LAN.
4 Explain why hub rooms are used in LANs.

1.3 Sending messages over the internet

In this lesson

You will learn:

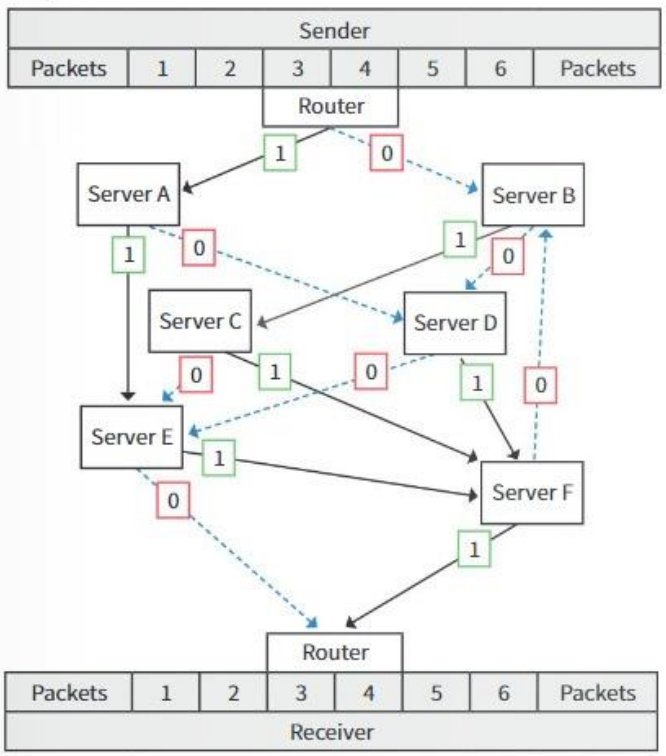
- the rules the internet uses to send messages.

Being connected

Being connected to the internet means you can send and receive messages. There are many types of messages you can send. These include simple text messages, audio, video and photographs.

To a computer, a text message and a video are treated the same – they are both **digital data**. A computer stores all files as digital data made up of zeros and ones. It sends every file across the internet in the same way.

In this lesson you are going to build a model. You will use it to learn how messages are sent over the internet. When you have completed the model, it will look like the one in the diagram. It looks complicated but you will build it step by step before using it in the next lesson.



Activity

Get ready to build your model. You will need a large sheet of paper, two or three coloured pencils or pens, and a ruler. If you don't have a large sheet of paper, stick two sheets of A4 printer paper together along the long edge.

First, add a sender section to the top of your sheet. There are three parts to it:

- the word 'Sender'
- a row labelled 'Packets'. This row should contain six boxes, numbered 1 to 6.
- a box labelled 'Router'.

Add a similar section at the bottom of your sheet. This section is called 'Receiver' and is a mirror image of the 'Sender' section you have drawn at the top of the page.

Do not draw the sections too large. Leave plenty of room in the centre of the sheet.

What are packets?

A computer does not send messages as a single section of data over the internet. Any file that you send over the internet is split into smaller sections before it is sent. The small sections are called **packets**. Whether the message is an email, a photograph or a word-processed file, it is split into small packets before being sent. This makes sending a message easier for the equipment in the network to handle. A packet is about the size of this paragraph (about 550 characters). If you could look inside a packet you would see digital data – 0s and 1s.

When the packets reach their destination, they must be put back together into the original message. You don't have to worry about creating packets or putting them back together. Your computer does it automatically whenever you send a message over the internet.

Many thousands of packets are sent if you are sending a photograph or another large message. Each packet is addressed to the person receiving the message. Each packet is numbered so that the message can be put back together in the right order.

What does a router do?

In Lesson 1.1, you learned that a router is used to connect your school or home network to the internet. Every packet in your message will pass through the router one after the other. When you use the model, you will move packets onto the router box before they are sent.

Activity

Go back to the model you started to draw. In the space in the centre of your page, do the following:

- Draw six boxes. Don't make the boxes too large. Make sure there is plenty of space between them.
- Label the boxes 'Server A' to 'Server F'. Look back at the diagram on page 14 if you're unsure how the boxes should look.

What does a server do?

As discussed in Lesson 1.1, the job of a server is to send files over a network. When you send a message over the internet, it is split into packets. Each packet travels in steps across the internet, jumping from one server to another until it reaches its destination.

Activity

In this activity you will add links, joining servers in your model together. In the real world, each server will be linked to many other servers. You will add just two links from each server in your model. This keep things simple.

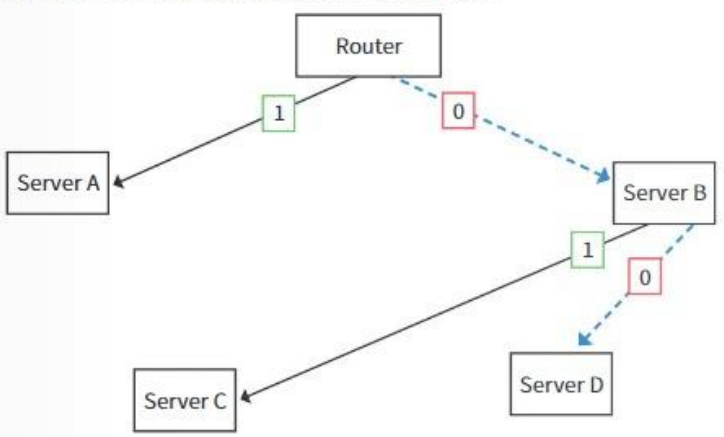
To make your model easy to follow, choose two different line styles.

- Use two different colours.
- Label one style of line '0' and the other '1'.

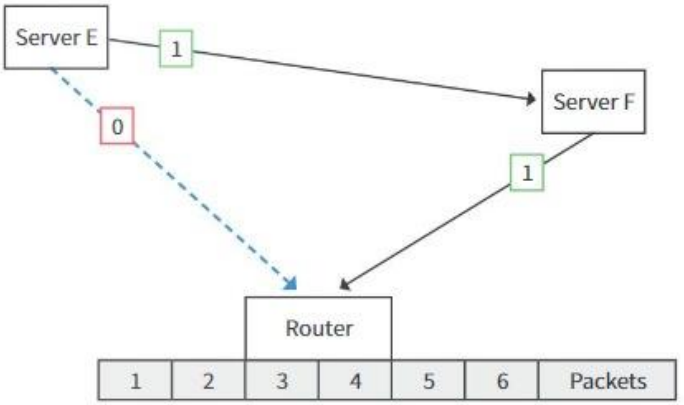
Add an arrow to indicate the direction of the line.

Start from the router at the top of the page. Draw connections to two servers. Use different line styles.

Work through each of the servers, adding two links to the other servers in your model. Make sure each server is linked to at least once.



Make sure two lines (one of each style) link to the router in the receiver, as shown in the image.



Extra challenge

Look back at the completed model from the start of this lesson. Check the model you have drawn to make sure it looks the same. If it doesn't, make sure you understand where you have made mistakes. Update your model.

How a packet travels over the internet

You have learned that a message is passed from server to server as it finds its destination. There is no set route for the message. When a server receives a packet, it must decide where to send it to next. The server will always send a message along the clearest route, avoiding slow or blocked routes.

Test

1 What is a router used for?
2 How does a packet travel across the internet?
3 Why are internet messages sent as packets?
4 Explain why packets sent over the internet have to be numbered.

1.4 Sending messages: a simulation

In this lesson

You will learn:

- how messages are sent over the internet using the packet switching method.

A simulation

In this lesson you will use the model you drew in Lesson 1.3 to show how messages are sent from one computer to another over the internet. The model shows how a message is sent step by step. You are going to use it in a simulation. A simulation copies something that happens in real life.

What you know so far

- All data sent over the internet is digital data.
- A message reaches its destination by being passed from server to server over the internet.
- A message is broken up into small packets before being sent over the internet.
- A router is used to connect a local computer to the internet. All messages sent over the internet will pass through the router.

Preparing for the simulation

You will need:

- your completed model
- a tracking table, which you will create in this lesson
- a coin
- six squares of paper, the same size as the numbered squares in the packets row in your model. Number the squares 1 to 6. These will be your packets of data.

Activity

Create a table like this.

Packet tracker								
	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Step 7	Step 8
Packet 1								
Packet 2								
Packet 3								
Packet 4								
Packet 5								
Packet 6								

Playing the simulation

Set up your simulation board. Start by putting the six numbered squares in the packets row of the 'Sender' section. The objective of the simulation is to move the message through the network of servers to the receiver section.

You are going to flip the coin during the simulation to decide which path each packet will take between servers. Decide which side of the coin represents '0' and which represents '1'. Make a note so that you remember.

Move Packet 1 to the router. Now you are ready to start.

Rules

- Flip the coin at the start of each round. The result will determine which path all packets will take in that round.
- Move each packet that is in play to the next server using the path (0 or 1) decided by the coin flip. A packet is 'in play' if it is on a server or waiting in the sender's router box. It is easier if you move the packets in number sequence from the lowest to the highest.
- In the table, record the server letter each packet has moved to. When a packet reaches the receiver router, write an 'R' in the table.

Packet tracker								
	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Step 7	Step 8
Packet 1	A	D	E	F	R			
Packet 2	B	D	F	R				
Packet 3	B	C	F					
Packet 4	A	E						
Packet 5	A							
Packet 6								

- Move any pieces that have arrived at the receiver router onto the next available slot in the packets row. Do not put the packets on their matching slot in the receiver's packet row. Put them on the next available slot in the order they arrive. So if Packet 6 arrives fourth in sequence, it goes on slot 4, not slot 6.
- Move the next sender packet onto the router.

Repeat the instructions until all packets have moved from the sender to the receiver. It will take around eight or nine rounds to complete the simulation. There will be up to three or four packets 'in play' at any time.



Activity

Play the simulation, then answer the questions below. You might want to play the simulation in a small group.

Look at the tracking table.

- Did all the packets take the same number of steps to reach the receiver?
- Did all the packets take the same route through the network?
- Did *any* of the packets take the same route through the network?
- Did the packets arrive with the receiver in the same order that they set off from the sender?

It is possible, in fact likely, that the packets you sent arrived out of sequence. There is a final step that needs to be added to the model to make sure that the receiver can understand the message. Do you know what that step is?

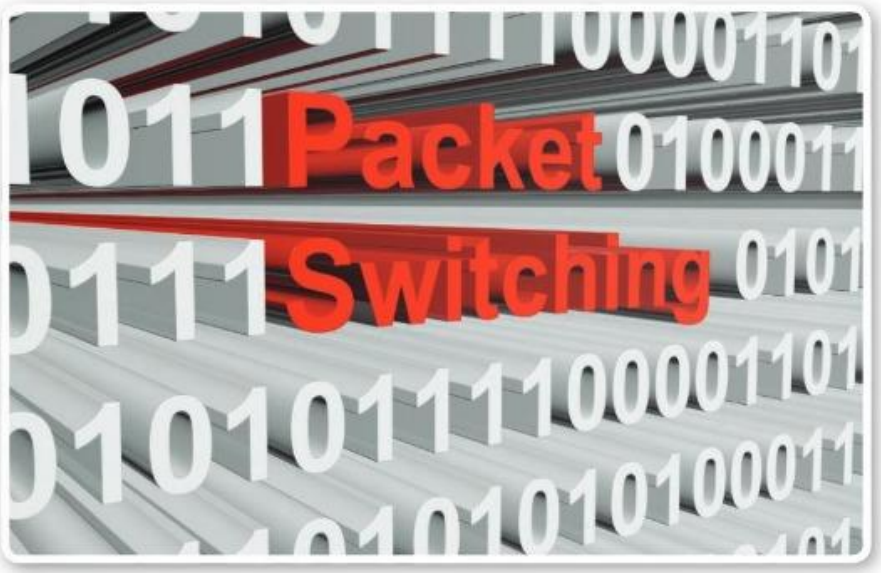
Packet switching

The simulation you have just played is a simple way of showing **packet switching**.

- A message is split into small sections called packets.
- As a packet travels across the internet, it is switched onto the clearest route.

Packet switching is a reliable way of sending messages over the internet. Almost all messages reach their destination correctly. Packet switching is used to send data across local area networks as well as the internet.

Packet switching is carried out automatically by your computer and other hardware in a network. You use packet switching every time you log on to a network.

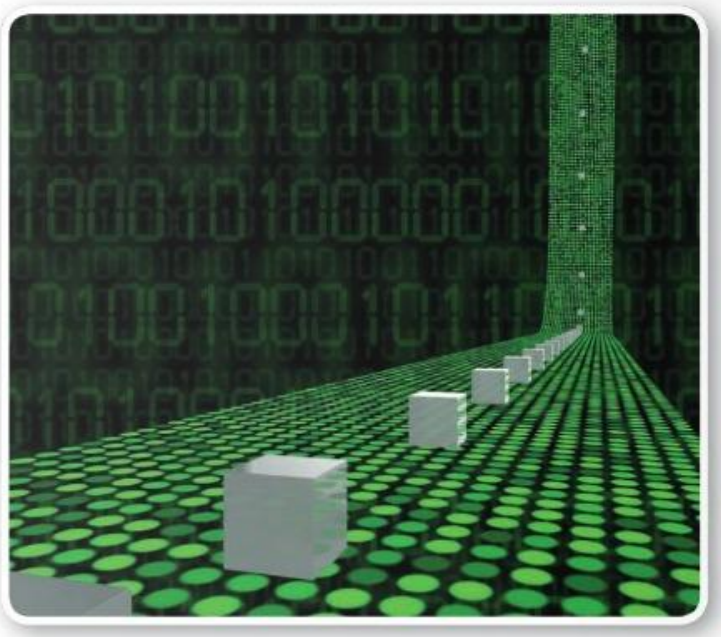


What a packet contains

You already know that a packet contains a small part of a complete message. The data in the packet is stored as digital data.

Each packet contains some extra data.

- The **address** of the person the message is being sent to. When you send an email, you must always add the address of the person you are sending it to (for example, a.friend@gmail.com). When your message is split into packets, each packet must contain the address.
- The **sequence number** of the packet. You saw in the simulation that packets can arrive at their destination in the wrong order. A sequence number allows the receiving computer to sort the packets into the right order.
- The **total number of packets** in the message. The receiving computer uses this information to check if it has received all the packets. If any packets are missing, the computer sends a message back to the sender and asks for them to be re-sent. This is done automatically so you don't have to re-send the whole message.



Extra challenge

Sometimes packets get lost as they are sent across the internet. What problem does that create for the computer receiving a message?

Research the web to find out how packet switching deals with lost packets.

Test

1. What happens if a packet gets lost when being sent across a network?
2. Name three pieces of information that are included in a packet when it is sent over a network.
3. Why can packets arrive at their destination in the wrong order?
4. Describe the term 'packet switching' in your own words.

1.5 Connecting to a network

In this lesson

You will learn:

- how to connect to a network
- basic troubleshooting methods for network connections.



Types of network connection

There are three main ways to connect to a network.

1 **Wired** Your computer is connected to a data point on a wall with a cable. You can only connect to a network if you have a network point nearby. You cannot move your computer. Wired connections are used for desktop computers and sometimes for laptops.

2 **Wireless (Wi-Fi)** A cable isn't needed for a wireless connection. Signals are sent between your computer and a wireless access point (WAP). Lesson 1.1 has more information about WAPs.

Laptop computers, tablets and smartphones connect to networks using Wi-Fi. Using Wi-Fi means you can move around without easily losing your connection.

Wireless networks are sometimes available in public places like restaurants or shops. A WAP in a public place is called a **hotspot**.

3 **Mobile phone** Smartphones can connect to the internet using a mobile phone connection. A connection can be made anywhere there is a phone signal.

How to connect to a network

School network

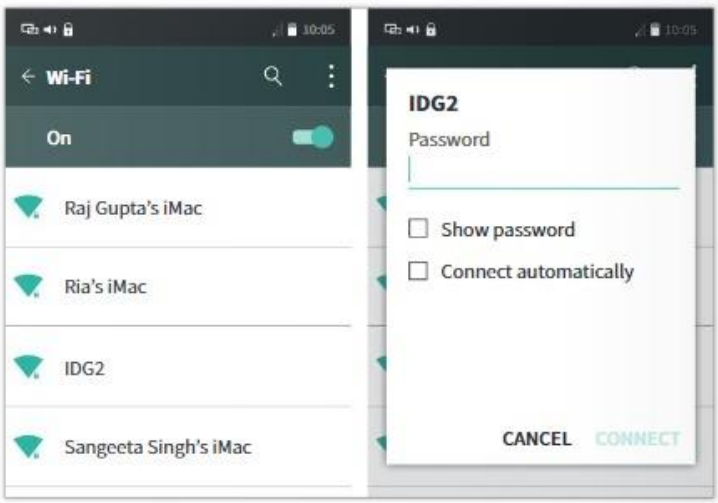
Everyone who is allowed to use the school network is given a username and password. You use the username and password to connect to the network. The login process is the same whether you connect by a wired or wireless connection.



Home network

You do not need a username and password for your home network. Everyone uses the same login details. To log in to the network, you need to know the network name and **network security key** for the router. This information is on a sticker on your router. The network security key works like a password.

When you switch on your computer, you will be shown a list of available networks. Your computer will detect your home network and any other networks close to you. It will detect your neighbour's network, if they are close enough, and any public hotspots.



When you click on the name of your network, you are asked to enter the network security key. Entering it correctly will connect you to the network. When you have entered the key once, your computer will remember it. If you tick the 'Connect automatically' box, you won't need to log on in future. Your computer will automatically connect to the network.

Solving network problems

When you are solving computer problems, you will be working with electrical equipment. Your priority is to stay safe. Always follow your school computer use safety rules. Always check with your parents before trying to deal with problems with your home network.

Stay safe

Always follow these safety rules.

- Switch off power points at the wall before connecting or disconnecting a mains electricity cable.
- Do not open the case of your computer or any other device.
- Never use force when connecting or disconnecting a cable or component.
- Make sure all cables are disconnected before attempting to move a device (for example, a printer).
- Do not lift heavy equipment on your own.
- If in doubt, STOP and ask for help.



Be creative

Create a poster to encourage people to act responsibly and safely when solving computer problems.

Make notes

When you are trying to solve network problems, record:

- the location of the computer
- the steps you have followed
- any error messages you see on-screen.

Username and password

If you are having problems logging on to a network, check you are using the correct username and password.

A password is case-sensitive. That means it matters if you use upper case letters instead of lower case ones. Check you haven't left 'Caps Lock' on.

If you have forgotten your password, use the 'Forgot password?' link to change your password. You will need an email address that a new password can be sent to. At school, you will need to ask your IT technician to change your password.

Wired networks

If you are having network problems with a wired connection, check that the cable connecting your computer to a network point is connected securely at both ends.

The connection socket on your computer will usually have a small green light. If the light is flashing, your computer is connected.

There are two things you can check if the light isn't flashing.

- The cable connecting your computer to the network may be faulty. Changing it for one you know is not faulty will solve the problem.
- The network point you are connected to may be faulty. Try another network point.



Only try to solve problems with your school network if your school IT policy allows it. Report the problem to the IT technician or your teacher and use another computer until the problem is fixed.

Wireless networks

If you cannot make a wireless connection to a network, check if you have wireless enabled on your computer. Find the Wi-Fi icon on the tool bar at the bottom of your screen. If it has a red cross on it, wireless on your computer may be turned off.

The picture on the right shows the wireless network symbol.



Click the Wi-Fi icon to open the wireless control box. You can turn wireless on by clicking the button marked 'Wi-Fi'.

If you are using a tablet or smartphone. you will find the same Wi-Fi icon on your device. If the icon is grey, wireless is turned off. Tap the icon to turn wireless on.

Home network

If you are having problems connecting to your home network, check:

- you are entering your password correctly
- that wireless is turned on
- whether other people in the house have a connection. If they do, the problem is likely to be with your device, not the router.

There are some basic checks you can carry out on your home network router.

- Is the router plugged into an electricity socket? Is the socket switched on?
- Check that any cables connected to the back of your router are secure.

If everything else fails, reset your router. Turn off the power switch or turn off the electricity socket. Wait 30 seconds then turn the router back on again. Do not press any button marked 'reset' on the router.



Activity

Write a list of instructions called 'How to connect to a network'.

Design the instructions for students who are about to start using computers, to help them study at school and at home.

Extra challenge

Work in a small group. Use your multimedia skills to create an audio or video guide to network connectivity. You can use a phone or recorder. Record yourself reading the instructions you wrote, or make a short video of someone connecting to the school network and explaining how to do it.

Test

1. List three ways to connect to a network.
2. What is a hotspot?
3. You are unable to connect your computer to your home network. List four problems that could be preventing you from connecting.
4. You want to connect a laptop to your school network. Why is a wireless connection better than a wired connection?

1.6 In the cloud

In this lesson

You will learn:

- about cloud storage
- about other services that are available in the cloud.

Spiral back

In Student Book 4 you learned about computer storage. You can save files using a storage device on your computer. Files can also be saved on a network storage device. You probably use network storage at school.

What is the cloud?

The **cloud** is a term used to describe the internet. The cloud is a metaphor for the internet. When you use a metaphor, you are saying that the internet is like a cloud, without using the word 'like'.

Saving a file to the cloud means saving the file to the internet.

Cloud storage

Everything you do 'in the cloud' involves data storage. When you save file in the cloud, it is stored on large storage devices. The storage devices are held in a **data centre**.

A data centre is a very large computer system. It contains many servers and storage devices. A data centre often has its own building.





Who provides cloud storage?

There are many providers of cloud storage. Your internet provider may offer cloud storage. Software providers like Microsoft and Google provide cloud storage. Microsoft's cloud storage is called OneDrive.

You often get free cloud storage when you buy a computer, or other devices like cameras and smartphones. Some companies specialise in providing cloud storage. Dropbox is an example.

Usually, a small amount of cloud storage space is provided free of charge. Businesses and people who have a lot of data can pay for more space if they need it.

Advantages of cloud storage

Here are the main advantages of cloud storage.

- You can access your files from anywhere you have an internet connection. If you save a file on your own computer, you can only access the file if you have the computer with you. If your file is in the cloud you can access it anywhere, using any computer.
- You can share your files with other people. For example, when you are working on team projects.
- Your work is safe. Cloud storage providers back up your files so that you don't have to do it. It is unlikely your data will be lost or stolen.

Other cloud services

Storage isn't the only reason for using the cloud. There are many other services.

Collaboration

As you saw in Student Book 7, collaboration means working with others.

Collaboration software makes working together easier. If files are stored in the cloud, any team member can work on them.

Collaboration software also lets people in different locations meet using web conferencing software. **Web conferencing** shows a video of the people taking part in the meeting. It also lets people show files on-screen and make notes. A web conference saves travel time for the people taking part.

Software applications

Many software applications (apps) work in the cloud, for example, Google Apps and Microsoft 365. The apps make it easy to work with other people. They make sharing documents and files easy.

Web hosting

If you want to build a website, you need a web server to store your web pages. A web server can be expensive, and most people do not have the skills to set up and run one.

Web hosting companies help people who want to set up a website. A web hosting service provides technical knowledge and web editing software. That makes creating a website easy.

Music, videos and games



Music and videos have been stored online for many years. When **digital** music was first stored online, people bought and downloaded the music. When downloading, you save a copy of the digital music file on your computer. When you have saved the copy, you can play the music on your computer, or transfer it onto a special MP3 player, such as an iPod.

Today, music and videos are often streamed to a computer or device like a smartphone. **Streaming** means that a music track is played as it is being downloaded to your computer. A copy of the file does not have to be saved before the music is played. Streaming is a cloud service.

Playing games in the cloud means that you can compete or collaborate with other players. This type of game is called a **multi-player game**. Multi-player games can be more interesting than single-player games. They often have social media tools included to allow players to chat about the game.

Advantages and disadvantages of cloud services

Here are the advantages.

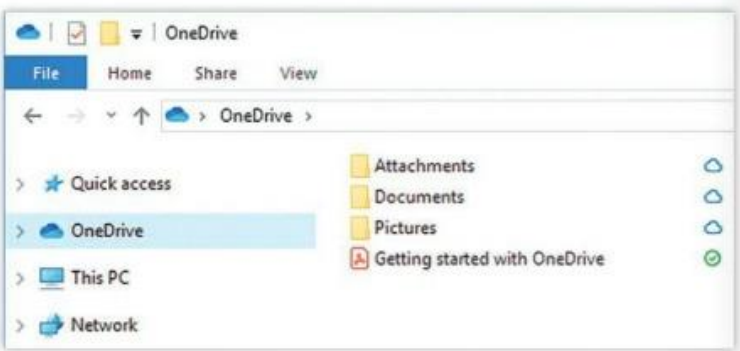
- Files are available anywhere and on any computer.
- Collaborating and sharing data is easy.
- Data is safe.
- Low cost to start with.
- Technical help is available from the company providing the service.

Here are the disadvantages.

- The service might not be exactly what you need.
- The owner of the service can change how it works.
- The owner of the service can increase costs.
- Can be expensive over time.

Saving to the cloud

When you save a file, you must choose where you save it. You can save it on a storage device on your computer. You can save it on a network storage device. You can save it to the cloud. The places will be listed for you when you save a file.



When you save a file to the cloud, a copy of the file is also saved to your own computer. This means you can open the file and work on it even if you have no internet connection.

You can choose not to make a copy of a file on your computer. The file will only be saved to the cloud. You will not be able to use the file if there is no internet connection. You can save large files like videos and photos to the cloud to save space on your computer.

Cloud storage uses icons to tell you how a file has been saved.

- A cloud status icon shows that the file is only saved in the cloud.
- A clock status icon tells you the file is saved to the cloud and your computer.
- An icon that looks like a person tells you the file has been shared with someone.



Activity

Your company is about to launch a new cloud data storage service.

Write an advert to attract as many new customers as possible. Make sure your advert highlights the advantages of cloud storage.

Extra challenge

Do you use any cloud services? For example, multi-player games, or streaming music and videos.

Make a list of the services you use. Which are your favourite services and why?

Is there anything you don't like about the cloud services you use?

Test

1. Describe three services that can be accessed in the cloud.
2. Where is data saved when it is saved to the cloud?
3. Explain why it is a good idea to store your data files in the cloud.
4. List three disadvantages of using cloud services.

Check what you know

You have learned

- about network hardware and how the components work together
- how messages are sent over the internet
- about packet switching
- how to connect to a network and how to solve problems with network connections
- about cloud storage and other services in the cloud.

Try the test and activities. They will help you to see how much you understand.

Test

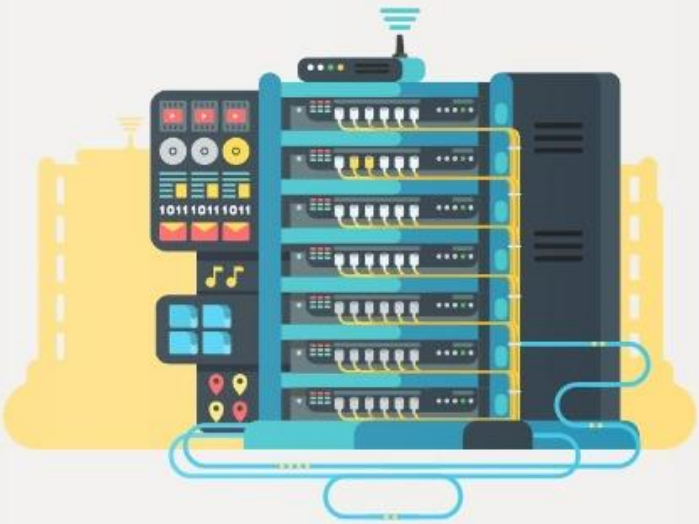
1. List two items of information you need to log on to a network.
2. Where is a data file stored if it is 'in the cloud'?
3. What are the two types of cable used to connect network equipment? Describe the advantages of each type.
4. Describe three computer services that can be used in the cloud.
5. You are having difficulties logging on to your home network. Describe three checks you can carry out to try to solve the problem.
6. Why can it be better to store data files in the cloud rather than on your own computer? Give three examples.



Activities

You have been asked to create a presentation. It will be used to explain to other students how the components of a network are connected to allow computers to communicate.

1. Create one or two slides that tell students how to connect to your school network and use the internet. You can include screenshots or pictures to help explain.
2. Create one or two slides about the items of network hardware you might find at home or in schools. You can include pictures or photos that you have taken yourself or found in internet research.
3. Add one or two slides that explain what to do if you have problems connecting to the school network. Give some hints about how to fix the problem.



Self-evaluation

- I answered test questions 1 and 2.
- I completed activity 1.
- I answered test questions 1–4.
- I completed activities 1 and 2.
- I answered all the test questions.
- I completed all the activities.

Re-read any parts of the unit that you feel unsure about. Try the test and activities again – can you do more this time?